Measuring emergence through collapse of the effective joint possibility space

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Abstract

Emergence is claimed for scaled-model abilities, multi-agent coordination and collective animal behaviour, yet each field certifies it with a different instrument, so identical behaviours can be labelled emergent or not depending on the measure. Here we define emergence as a spontaneous, persistent, regime-level collapse of a system's effective joint state–action–trajectory possibility space, decomposed by source into environmental, individual, pairwise and higher-order channels. Calibrated on analytic ground truth and validated on off-design physics, the instrument shows that learning populations commit abruptly wherever a joint exploration barrier—a convention or role with value only once shared—must be crossed, and smoothly otherwise; that a capability can form smoothly across training while each deployment of it commits abruptly within an episode; and that remaining openness predicts, prospectively and causally in a standard benchmark, whether an intervention can still change the outcome. Possibility-space collapse turns a contested word into a measured, falsifiable, actionable quantity.

Introduction

"More is different" is a claim about organization: many interacting parts settle into a collective regime none of them specified in advance[1–4]. The word *emergence* names that phenomenon in statistical physics[5,6], in animal collectives[7–10], in the new capabilities of scaled neural networks[11,12], and in multi-agent reinforcement learning[13–15]. Yet the fields do not share an instrument: causal-emergence measures ask whether a coarse-graining has more causal power than its microstate[16–19]; autonomy measures test macro-variable self-determination[20,21]; partial-information decomposition asks how much of a target is carried synergistically[22,23]; phase-transition theory tracks an order parameter[24]; complexity science invokes irreducibility[25,26]; and machine learning reports "emergent abilities" as sharp benchmark upturns against scale[11,27,28], a debate entangled with scaling laws[29,30] and grokking[31,32]. Each family evaluates a chosen description—the causal power of a coarse-graining, the synergy of a target variable, the trajectory of a hand-picked order parameter; none offers a system-agnostic test for *when* a collective regime forms, *how abruptly*, and from *which* interaction structure.

What the fields share is not a measure but an intuition: an emergent phenomenon is unexpected beforehand yet lawful in retrospect—surprise has even been proposed as its defining test[33]. But surprise is an observer property, and a criterion that anything unfamiliar can satisfy names everything and measures nothing. The formal measures fare little better: some reported emergent abilities are metric artefacts—change the score function and the sharp jump disappears[34]. This is the generic risk of defining emergence on the *observable of convenience*—a benchmark, an order parameter, a success rate—rather than on the object that actually reorganizes. Worse, two systems can produce byte-identical outcomes—identical joint distributions, marginals and task success—through entirely different couplings (Extended Data Fig. 1a), and no statistic of that shared distribution can tell them apart. They become distinguishable only relative to a declared *observation contract*—which variables are measured, which are exogenous, which counterfactuals are admissible—under which one can ask how the effective possibility space closes.

The object is the joint space of what every part *could* still do together—the effective joint state–action–trajectory space. Before organization it is wide: an ant colony approaching a gap can still bridge, cross elsewhere or disperse[35]. Emergence, on our account, is the moment this space *collapses* onto a narrow committed regime—the colony decides, together, to build—and collapse is not completion: the bridge does not yet exist when the possibility space has contracted onto building it (Fig. 1). This definition keeps both halves of the surprise intuition and makes each measurable: before commitment, low-order observables carry no warning of what is coming; after it, the collapse has reproducible timing, sources and consequences. Verdicts are stable within a measured admissible equivalence class of contracts (Methods).

Here we make three contributions. First, a two-level definition: a qualification (spontaneous, regime-level, persistent) and a quantitative *emergence intensity profile* (EIP) whose axes—amplitude, abruptness, source decomposition—are separately measured, so abruptness is a phenotype, not a smuggled existence criterion. Second, a calibrated measurement contract, matured through preregistered self-falsification, that recovers a collapse's source on analytic ground truth (72/72 cells) and on off-design physics. Third, the finding that in *learned* systems the collapse dissociates cleanly by timescale, varies systematically with system size and coupling, predicts controllability, and is **not** generic: the punctuated form arises spontaneously where learning must cross a joint exploration barrier, while ordinary dense-reward optimization collapses smoothly.

Results

An operational definition on the right object

We measure openness as the normalized entropy of the effective joint possibility distribution, $O_t = H(P_t)/H(P_{\text{ref}})$, and collapse as $C_t = 1 - O_t$. Because literal support never shrinks for a stochastic policy, we work with the *effective* possibility count $N_{\text{eff}} = 2^H$—the formal version of the $10^n \to 2^n$ intuition. A system *qualifies* as emergent when its collapse is regime-level (a structural change in the joint distribution, not a marginal one), endogenous (generated internally, declared through a provenance boundary rather than assumed), and persistent (stable over the declared horizon; recovery and causal irreversibility are reported separately as robustness properties). Given qualification, we report an intensity profile rather than a binary label. P_t itself is defined relative to a declared *observation contract*—variables, horizon, discretization, reference, provenance boundary—frozen before any adjudication; objectivity enters through freezing, measured invariance and refusal outside the contract's domain (Methods).

The decisive move is the source decomposition. Total collapse splits along a nested maximum-entropy ladder,

$$C_{\text{total}} = C_{\text{env}} + C_{\text{individual}} + C_{\text{pair}} + C_{\text{high}}, \tag{1}$$

so that the interaction cut is a *decomposer*, not a gate, in the spirit of connected-correlation expansions[36]. On analytic ground truth with four independent knobs, each knob moves only its own component, the ladder is monotone, and hiding an environmental common cause deterministically re-attributes collapse from C_{env} to the relational channels (five preregistered checks pass; Extended Data Fig. 1c)—contract-relativity made precise: "higher-order" is defined only relative to what an observer declares exogenous. On a 72-cell factorial (4 sources $\times$ 3 temporal shapes $\times$ 2 stabilities $\times$ 3 magnitudes) the instrument recovers the source in 72/72 cells, and—the point that defends against the metric-artefact critique[34]—amplitude M is invariant across temporal shape (relative range 0.000) while abruptness J strictly orders punctuated $>$ sigmoid $>$ gradual (Extended Data Fig. 1b); a revelation-only or discontinuous-metric control produces exactly zero collapse.

A breakpoint marks a phenotype, not emergence itself

Qualification distinguishes emergent from non-emergent organization; the *breakpoint* distinguishes the punctuated profile from gradual, non-monotone or unresolved collapse—so a missing breakpoint never by itself rules emergence out. We fit the collapse curve with a two-regime model against a one-regime fit by ΔBIC, inside a saturation-truncated window and behind an effect-size gate—two amendments forced by preregistered misses and fixed definitionally on fresh seeds. The matured detector was frozen and scored on a held-out labelled benchmark: zero false positives across deceleration, gradual and flat control families at every tested grid density and noise level, onset power 1.00 at our operating resolution, and location agreement with a standard change-point method (Methods). Every headline verdict below is additionally stable across 243 adjudication contracts (gate, window, truncation, threshold, grid) and—separately—across a preregistered battery of representations of the openness object itself; the measured equivalence class, including the representations that break it, is reported in Methods.

With the matured detector, an *onset*-type breakpoint (slow $\rightarrow$ fast) follows the preregistered profile predictions—present where the theory predicts punctuated commitment, absent where it predicts smooth convergence. It appears in a committing ant colony, in a learned high-order coordination task (7/8 seeds across original run and five-seed extension), and at the Kuramoto synchronization transition[24,37]. It is absent—correctly—in an ordinary supervised learner, whose collapse begins at maximum rate and decelerates (a *knee*, not an onset), and unresolvable—so not claimed—on stored language-model checkpoints, whose entropy collapses before the second saved step.

In learned systems, formation and realization dissociate

The most consequential results are in learned agents, where the possibility space is shaped by optimization rather than fixed physics. In a two-phase collective-transport task—sixteen agents must jointly grip an object before they can push it, structurally delaying the choice of side—policy-gradient training[38] learns in 10/10 seeds (success ≥ 0.995), all with the predicted signature: side-openness holds at a plateau for $\sim$17–19 steps and then collapses, with an onset breakpoint in every seed (ΔBIC 45.8–52.7, t^* at 16–18; Fig. 2b,d). A preregistered mechanism control—the same task without the gripping phase—

shows no breakpoint (0/5): it is the structural delay, not learning per se, that produces punctuated collapse (Fig. 2c). The phenomenon reproduces under a different algorithm[39] (advantage actor–critic, byte-identical environment): 5/5 seeds reproduce the plateau-then-collapse shape and primary hinge (ΔBIC 37.7–45.5)—a partial replication: the strict thinning clause passes in only 3/5, missing solely on thinned-subsample detector power.

The same learned system cleanly separates the two senses of emergence that are usually conflated (Fig. 3). Along the *realization* axis—within a single episode—collapse is punctuated (5/5 breakpoints). Along the *formation* axis—across training—the capability appears *smoothly*: outcome-openness expands ($0 \to \sim 0.65$ within ~ 100 updates) with no structural onset at either standard or fine 5-update resolution (0/5 on both). Formation is smooth and expansive; realization is punctuated. This dissociation, in one system with one instrument, clarifies why debates over whether scaled-model abilities appear "suddenly" have been inconclusive[11,34]: the capability may form smoothly while each deployment of it commits abruptly.

Punctuated collapse arises without designed barriers

A fair objection is that the grip task's delayed commitment is imposed by its own state machine—a positive control, not a discovery about learning. We therefore preregistered two systems in which *no* environment mechanism delays commitment: every joint regime is available, and exactly equivalent, from the first update. In a ten-agent signalling population (five meanings, five symbols, reward only for successful decoding[40]), any of the 120 codes could be adopted at any moment; yet convention openness holds at a statistically flat plateau and then collapses with an onset breakpoint in 4/5 seeds (ΔBIC 17.8–30.1), the breakpoint ($t^* = 275$–300) preceding mutual intelligibility 0.9 (updates 700–1,025) in every onset seed, and the five seeds converging to five *different* codes. A six-agent division-of-labour task (team reward only when all six interchangeable roles are covered) shows the same signature more strongly in 5/5 seeds (ΔBIC 53.6–71.7; closing slope $\sim 30\times$ the plateau slope), again with collapse preceding capability and five distinct role permutations. In both systems the open plateau *is* the joint exploration barrier: mid-plateau, a probe agent unilaterally committing to the *best* available code or role gains essentially nothing (0.011; 0.001); after commitment, the cheapest unilateral deviation forfeits the structural maximum of the payoff (0.40 of a 0.40 ceiling; 1.00); and populations from different seeds are mutually incompatible (cross-seed intelligibility 0.14, hybrid-team success 0.05, versus ~ 1.00 within). A code or role division has value only insofar as others already share it, and the collapse is nucleated, self-amplifying commitment—the learned analogue of sharp consensus transitions in naming games[41,42]. Neither system depends on its tabular parameterization: randomly initialized neural policies (per-agent MLPs for signalling; one *shared* MLP for all six role agents) reproduce the signature—onset in 6/7 learned convention seeds and 10/10 role seeds (ΔBIC up to 162), collapse preceding capability in every one, every seed a different code or permutation. The neural runs expose a mechanism the theory predicts: random initialization pre-breaks the symmetry among equivalent regimes, compressing the plateau roughly tenfold (Methods), and an initialization-scale sweep moves commitment time in the predicted direction (monotone for signalling; role-system midpoint non-monotone, a registered miss). This mechanism—seed randomness deciding when and where a run commits—matches the finding that emergent language-model capabilities reflect bimodal cross-seed distributions rather than deterministic scale thresholds[43]: each seed crosses the same barrier at a stochastic time, so ensemble averages blur what single-run collapse curves resolve. Punctuated collapse is therefore no artefact of designed gates or tabular parameterization: it arose in every tested system where learning must cross an endogenous joint-regime barrier, with barrier width set by how symmetric the competition starts.

The source typology transfers to learned coordination

Each channel of the ladder is realizable by learning (Fig. 4). A hidden-coordination task with eight individually-observing agents produces a purely *relational* collapse: per-agent marginals stay open (individual entropy ≈ 1.0 bit) while total correlation rises to 6.8/7 bits (Fig. 4a). When the low-order route is blocked with private information, learning builds a genuinely irreducible *higher-order* carrier: the textbook XOR structure emerges with C_{high} 0.94–0.96 bits and pairwise ≈ 0.0004 bits throughout the whole formation history (3/3 seeds; Fig. 4b), and declaring the private cues exogenous re-attributes the identical collapse to C_{env} (Fig. 4c)—the learned version of contract-relativity. Together these give a sharp statement: higher-order structure is not unlearnable but *unfavoured*—gradient learning selects the lowest-order implementation available and builds irreducible structure only when information constraints force it[44], echoing difficulties in emergent-communication[45–47] and cooperative-MARL[48–50] studies. Off design, the same ladder correctly reads out Kuramoto oscillators (uncoupled $\rightarrow$ null; common driver $\rightarrow$ C_{env}; single edge $\rightarrow$ C_{pair}), a mechanism vocabulary it never saw. And at matched task performance, the source *profile* separates a learned Overcooked pair[51] from a noise-matched scripted pair through its environment-conditioned component (C_{env} 0.0137 [0.0123, 0.0156] vs 0.0005 [0.0004, 0.0007]; Fig. 4d), even though a single-point coupling measure cannot—the difference is visible in the collapse composition though in no scalar summary.

Remaining openness predicts controllability

If emergence is the closing of a possibility space, then how open that space still is should predict whether the outcome can still be steered. It does (Fig. 5). In the ant system, per-episode flip-rate rises strictly with the episode's own remaining openness ($0.000 \rightarrow 0.205$ across bins; closed episodes flip 0/2,600; flipped episodes are 0.58 openness units more open, permutation $p < 10^{-4}$ over 8,372 paired counterfactuals; Fig. 5d). In the learned grip system, a counter-regime impulse switches the outcome with probability 1.0 up to $t = 16$ and only 0.27 by $t = 30$, and side-openness predicts switchability with AUC 0.996 (Fig. 5a,b). This is not a time proxy: at fixed intervention time, openness still separates flippable from locked episodes (AUC 0.974–0.990 at every tested τ; 20,480 episodes per cell). Conditioning additionally on the physical order parameter removes the signal—the registered boundary: here openness is an information-sufficient transform of the physical state, and its value is that it is computable from the policy alone and transfers as a decision rule. It also works *prospectively*: a rule that waits until openness first drops to a threshold calibrated on the original seeds transfers to five unseen seeds, flipping 99.99% of episodes while acting 3.8 steps later than the best fixed schedule (99.6%) and beating random timing by 21 points—a policy-accessible intervention coordinate, acting as late as each episode allows. A matched-parameter contrast isolates the mechanism: when stances are inertial (a hidden consolidation phase exists), joint openness predicts switchability *better* than the physical order parameter (AUC 0.886 vs 0.849);

when the single stickiness parameter removes the hidden phase—all other constants unchanged—the advantage reverses (0.811 vs 0.884; Fig. 5c). Both orderings were preregistered predictions. Openness carries controllability information beyond the order parameter exactly when there is a hidden regime to consolidate.

Abruptness scales with system size and criticality

The breakpoint is not specific to learning; it is a systematic feature of collective self-amplification. In an ant-commitment model a single chooser *never* shows onset at any gain or grid, while large colonies always do, and the dependence is quantitative (Fig. 6a). Because commitment nucleates from binomial fluctuations of relative size $N^{-1/2}$ that are then amplified at an N-independent rate—a derivation recorded before running—commitment time should grow logarithmically while the collapse profile stays fixed. Both predictions hold across $N = 1$–500: $t_{50} = a + b \ln N$ with $b = 87.6$ (bootstrap 95% CI [41, 124], $R^2 = 0.93$); the closing width is N-invariant (280–290 trips from N=50 to 500); and the median collapse curves for $N \geq 50$ fall onto a single universal profile under pure time translation (mean pairwise RMS 0.010 versus 0.266 unaligned)—sharper than equilibrium-style width rescaling[5]. One sub-prediction failed and is reported: at the longer horizon the onset threshold sits between $N = 10$ and 20, so "onset at every $N \geq 10$" was counted a miss. In the Kuramoto system the same instrument tracks the approach to criticality: across coupling $K = 0.9 \to 2.5$ (10/10 onsets) the breakpoint time falls monotonically ($6.7 \to 1.8$) and the closing slope rises monotonically ($0.032 \to 0.199$), every subcritical run is correctly gated null, and the collapse is carried by the pairwise channel (3/3 seeds; Fig. 6b)[52]. Abruptness therefore has two independent, manipulable control parameters—system size and feedback strength—both preregistered and confirmed.

Where the punctuated form does not appear

Onset-type collapse is **not** generic to learned optimization (Fig. 6c): in standard two-agent Overcooked the learned policies collapse gradually on both a policy-entropy and a trajectory-occupancy object (0/3); a registered rescue sequence—smooth consensus, quorum populations, learning-rate scans—also produced no onset (0/20).

The theory names the missing ingredient—cramped-room geometry pins the efficient plan, leaving no competing joint regimes—and restoring it *within the same standard package* restores commitment. In the official coordination-ring layout, clockwise and counterclockwise circulation are mirror-equivalent conventions with value only when shared: with identical PPO mechanics, all eight preregistered self-play seeds end committed to one direction (final direction probability 0.97 or 0.03; six counterclockwise, two clockwise—endogenous symmetry breaking in an unmodified benchmark), while neither cramped-room control ever commits. The frozen detector certifies a punctuated onset in one seed (breakpoint at 780k steps preceding the capability crossing at 1M) and declines the rest: their formation is non-monotone—commitment forms, dissolves and re-forms—which the monotone-collapse instrument was not built to certify, so we count the onset-rate clause as a miss. The regime-level fact surviving every seed is the committed final convention. A preregistered within-episode probe locates the commitment level: mid-training episodes drift with a direction bias but never internally lock (0/5), whereas final-checkpoint episodes are closed from the first step (initial openness 0.37, 8/8 seeds; untrained controls stay open, zero false onsets). The ring convention is a formation-level commitment with no realization stage—the mechanistic complement of the grip system, whose commitment lives within the episode.

A final preregistered experiment tests whether this commitment is causally real. We added unbiased Gaussian noise (two strengths) to checkpoints taken very early, at the most open point and well after commitment, then resumed training for 400k steps with byte-identical mechanics (8 seeds, 48 continuation runs; the seed is the independent unit). The preregistered primary endpoint—the perturbation flips or decommits the final convention—occurs in 7/8 seeds perturbed while open versus 1/8 after commitment (exact paired sign-flip $p = 0.016$; run-level 8/16 versus 1/16, Fisher $p = 0.008$). Strict direction reversal is secondary and descriptive (three open-phase flips, zero late; $p = 0.13$ across seeds). Stored openness at the perturbed checkpoint predicts strict flips (AUC 0.85; descriptive, runs cluster within seeds), and capability recovers to 0.92 of the unperturbed rate after late perturbation: what locks is the institution, not the learning. This validates the functional open-versus-committed distinction, not a causal breakpoint time (open and late checkpoints also differ in training time; Methods).

Among classic cases the picture is mixed: Vicsek flocking[53], Swift–Hohenberg pattern selection[54] and Schelling segregation[55] organize strongly but collapse *gradually*, while the Potts model[56] reproduces the continuous-versus-first-order distinction through hinge strength and hysteresis. The result is a classification: collapse profiles sort systems into punctuated, gradual, parameter-axis and early-warning forms, and onset appears specifically where a joint-regime barrier—physical (the grip gate), informational (the blocked low-order channel) or endogenous to learning (a convention or role with value only once shared)—must be crossed. Dense-reward optimization with no such barrier collapses gradually, and our instrument says so.

Discussion

We have argued, and measured, that emergence can be defined on the object that actually reorganizes—the effective joint possibility space—and that this choice avoids the failure mode behind the metric-artefact problem[34]: amplitude and abruptness are provably separate axes, and a discontinuous scoring function produces zero collapse. The definition is generative rather than nominal: it supplies a source typology that transfers from analytic ground truth to physics to learned agents, a breakpoint whose presence and absence track the theory's predictions, manipulable dependences of abruptness on size and coupling, and a control-theoretic payoff: remaining openness predicts whether an intervention can still flip the outcome.

For machine intelligence specifically, the formation/realization dissociation clarifies a persistent confusion. "Emergent abilities" of scaled models[11] and the within-run appearance of a coordinated strategy in self-play agents[13,57,58] are different events on different timescales: our learned flagship shows the first as smooth and expansive and the second as punctuated. Concurrent work detects higher-order structure in multi-agent language-model groups through partial information decomposition[59]; that programme asks whether synergy is *present*, ours when the joint possibility space *closes*, how abruptly, through which channel, and whether the closure can still be steered.

Onset-type collapse is not generic to deep MARL at current scale: dense-reward Overcooked optimization collapses gradually. Our designed systems (the grip gate; the blocked-channel XOR task) are mechanism-isolating controls, not load-bearing ones: convention formation and role lock-in reproduce the signature with no designed barrier. What separates the two classes, on present evidence, is a joint exploration barrier—now directly measured—and the coordination-ring experiment tests this boundary inside the very benchmark that produced the negative: restoring competing regimes restores committed conventions in every seed. Certified onsets remain rare there because that commitment is non-monotone—an instrument boundary (a settled-envelope extension failed its own validation and was retired; Methods); certifying non-monotone commitment and extending to large-scale self-play are the clear next steps. Three boundaries remain: the commitment window is not causally verified (a one-pulse hypothesis failed its falsification clause; formation proved re-entrant); stored language-model checkpoints are too sparse for any breakpoint claim; and representation-independence holds as a measured equivalence class, breaking only where a representation erases the latent channel. Nor is "emergence = onset breakpoint" a universal equivalence: several canonical cases collapse gradually. What we offer is narrower and more useful—a calibrated, falsifiable instrument that measures *how*, *when* and *through which channel* a possibility space closes, and that reports every place it says no. A natural next step is to connect the collapse object to internal-representation geometry in trained networks[60], extending the instrument from behaviour to mechanism.

Methods

Possibility space and openness. For a system of n parts we define the effective joint possibility distribution P_t over the relevant state–action–trajectory support at analysis time t, estimated from rollouts or exact enumeration where tractable. Openness is $O_t = H(P_t)/H(P_{\mathrm{ref}})$ with H the Shannon entropy in bits and P_{ref} the maximum-entropy reference on the same support; collapse is $C_t = 1 - O_t$ and effective possibility count is $N_{\mathrm{eff}} = 2^{H(P_t)}$. Entropies are estimated with a Miller–Madow-style bias correction and, for the joint objects, with the nested ladder below rather than by direct high-dimensional estimation.

Source decomposition. The nested maximum-entropy ladder constrains, in order: (i) the statistics of the declared exogenous variables and conditional independence given them (environment-conditioned); (ii) additionally each part's marginal (individual); (iii) additionally all pairwise marginals, following the connected-correlation construction of Schneidman et al.[36] (pairwise); and (iv) the full joint. C_{env}, $C_{\mathrm{individual}}$, C_{pair} and C_{high} are the successive entropy reductions. Three properties matter for interpretation. First, each stage only adds constraints, so entropy is non-increasing along the ladder and every component is non-negative by construction. Second, the ladder is a sequence of information projections defined for *every* joint distribution—it is not a parametric model of the data—so "model misspecification" does not arise; a generator that mixes sources simply deposits entropy reduction at several levels at once, which is the intended reading. Third, the stage order is not a free knob: given a declared exogeneity boundary, the individual-to-pairwise-to-higher progression is the canonical hierarchy of marginal constraints, and the only genuinely movable choice—the boundary itself—is exactly the declared contract-relativity demonstrated in Fig. 4c. The provenance boundary is fixed before analysis in every experiment. A preregistered stress audit quantifies the remaining freedoms. Off-family generators the ladder was never built from are attributed correctly (a modular-sum triple with exactly independent pairwise marginals loads 3.32 bits on C_{high} with $C_{\mathrm{pair}} = 0$; a Markov copy chain loads purely on C_{pair} with $C_{\mathrm{high}} = 0$; fifty random Dirichlet joints give zero negative components and an exact telescoping identity). The constraint sets form a filtration, so no individual/environment reordering exists; the one genuine order freedom—declaring the environment before or after the pairwise stage—leaves the individual and higher-order components exactly invariant and moves the environment/pairwise split by at most 0.26 bits across an 81-cell grid. Finite-sample behaviour is measured, not assumed: median absolute component error is $\leq$0.018 bits at 3×10^4 samples for the three-agent test system, with no negative estimates in 120 runs, and the small-sample bias concentrates in C_{high} (0.31 bits at $n = 300$), which sets an explicit sample floor for higher-order claims. One registered audit prediction missed: when generator knobs interact strongly (pairwise copying under a parity mechanism), realized structure genuinely relocates across orders—copying makes the parity constraint first-order—so the ladder attributes it to the individual level. We report this as a property, not a defect: the ladder measures realized distributional structure, not generator labels.

Breakpoint detector. Collapse curves are fit with one- and two-regime piecewise-linear models; the breakpoint is accepted when the two-regime ΔBIC exceeds 10 and survives parity-thinning of the checkpoint grid, the curve passes an effect-size gate (max drop ≥ 0.1), and the analysis window is truncated at saturation (first point within 5% of the final value). An *onset* type requires $|\text{slope}_{\text{after}}| > |\text{slope}_{\text{before}}|$; the reverse ordering is a *deceleration* knee. The gate and saturation-truncation were added in response to two preregistered control failures (a flat subcritical false positive; a saturation-knee capture) and re-tested on fresh seeds.

Detector validation. Because the detector was matured through its own registered failures, it was frozen and then scored on an independent, preregistered synthetic benchmark of 200 curves per family (onset, deceleration knee, linear gradual, sigmoid, flat) per condition, with labels fixed before running. At the reference operating point (80 grid points, noise $\sigma = 0.02$) onset power is 1.00 and the false-positive rate across all control families is 0.000; power rises monotonically with grid density (0.00/0.62/0.995/1.00 at 12/20/40/80 points) while the false-positive rate remains 0.000 at every density, and degrades gracefully with noise (0.93 at $\sigma = 0.08$, false positives 0.000 throughout; Supplementary Table 1). Zero observed false positives among 200 curves per control family bounds the true rate at $\leq$0.5% per condition (95% rule-of-three over the pooled 600 controls); we report the bound rather than claiming the rate is exactly zero. The detector adjudicates monotone collapse only: a preregistered extension to non-monotone commitment (a settled-openness envelope recording irrevocable closure) kept false positives below 0.05 in every validation round but never exceeded 0.71 power against its required 0.90 on synthetic non-monotone positives, and under its own preregistered stopping rule was closed without ever being applied to system data; certifying non-monotone regime formation therefore remains an open instrument problem, which is why the coordination-ring onset count is reported at 1/8. A semi-synthetic injection audit quantifies the matching noise floor: commitments injected at known times into the ring pipeline's own estimator (real grids, real committed-episode counts, binomial direction noise) are recovered with false-positive rate 0.01 and median t^* error 1% of span, but detection power saturates at 0.88 (registered threshold 0.90, recorded as a near-miss)—the 30-episode evaluation budget itself caps single-curve power below the synthetic-library ceiling. Breakpoint locations agree with an independent standard change-point method (binary segmentation, RBF cost) to within 10% of span on onset curves; unlike generic change-point detection, the instrument correctly declines to call onset on deceleration and gradual curves. The zero power at 12 grid points sets the resolution floor: it is why no claim is made on sparse language-model checkpoint grids.

Adjudication-contract robustness. All headline breakpoint verdicts were re-adjudicated across 243 analysis contracts per system, crossing saturation fraction $\{0.02, 0.05, 0.1\}$, analysis window $\{0.75, 0.875, 1.0\}$ of the curve, effect-size gate $\{0.05, 0.1, 0.15\}$, ΔBIC threshold $\{8, 10, 12\}$ and checkpoint-grid stride $\{1, 2, 3\}$. For the grip flagship, the committing ant colony and the learned high-order task the onset verdict is unchanged in 100% of adequate-resolution cells and the breakpoint location varies by 1%, 8% and 11% of curve span respectively; aggressive grid coarsening reduces detectability (a power effect quantified by the validation benchmark above), never the location (Supplementary Table 2). The collapse object itself is fixed by the theory (the joint commitment space), not chosen by fit: the raw action-entropy of the grip system *rises* during the same window and is reported as such.

Observation contract. Every entropy is computed relative to a declared contract with six components: the variables admitted to the joint space, the trajectory horizon, the discretization, the reference distribution (maximum entropy on the same support), the probability-truncation rule, and the provenance boundary separating environment from system. Two properties anchor the contract's objectivity. First, an invariance that holds exactly: O_t is unchanged under any bijective recoding of states or actions, because Shannon entropy is permutation-invariant and the reference is defined on the same support—openness cannot be manufactured or destroyed by renaming. Second, an admissible class that is measured rather than assumed: representations remain admissible while they separate the competing regimes at or above the detector's resolution floor, and the equivalence battery below maps where that class ends (non-injective coarse-graining that merges regime-distinguishing states, or truncation that deletes the latent channel, breaks the verdict—mechanistically, not capriciously). The contract is frozen before adjudication and the instrument refuses outside it. One component deserves emphasis: the framework does not discover the regime variable from raw trajectories; it certifies a *declared* regime object. In every system here that object was named before analysis with a domain justification—side choice (bridge), grip configuration and push side (transport), code mapping (signalling), role permutation (division of labour), XOR carrier (blocked channel), circulation direction (Overcooked ring), phase locking (Kuramoto), magnetization sector (Potts)—and where no credible regime object exists the correct output is refusal, not a verdict. The instrument is thus a general measurement contract for declared collective regime objects, not an automatic emergence detector for arbitrary raw systems. The analyst's freedom is thereby the same freedom every measurement has—the choice of ensemble in thermodynamics, of coordinates in mechanics—made explicit, fixed in advance, and accompanied by its measured range of validity.

Representation equivalence. Detector settings and measurement representations are different freedoms, so they were audited separately. On byte-identical retrained seeds we recomputed the openness object under twelve preregistered representations—for the convention system, the population-mean mapping, per-agent mean entropy, the listener-side dual, a 2,048-sample behavioural estimate, a checkpoint-zero empirical reference, $\varepsilon = 0.01$ truncation and 5$\rightarrow$3 symbol coarse-graining; for the grip system, policy probabilities, state quantization at 0.1 and at 0.5, realized 16-agent action counts, and $\varepsilon = 0.01$ truncation—and adjudicated each with the frozen detector. The onset verdict survives in 8/12 cells and the breakpoint location is essentially representation-invariant wherever a verdict exists: $t^* = 250$–300 of a 4,000-update span (1.25%) in the convention system, and identical ($t^* = 17$) in every preserving grip representation. The registered $\geq$90% preservation bar was missed, so per the registered clause we report the measured equivalence class instead of claiming blanket invariance. All four breaking cells fail for identifiable reasons and still locate the commitment at $t \approx 19$–22: two fall just below the conservative ΔBIC$= 10$ bar while agreeing on location (per-agent entropy, which is an individual-level object rather than the joint convention; 5$\rightarrow$3 binning); 0.5-level state quantization compresses the transition into a one-to-two-checkpoint cliff that fails the frozen thinning clause (a resolution limit, not a disappearance: the unthinned fit gives ΔBIC 111, onset-type); and ε-truncation zeroes the side channel while it is latent ($<$1% of action mass during the grip phase), making the object non-monotone—truncating small probabilities destroys precisely the still-open possibilities the object exists to measure. The equivalence class is therefore: verdict and location survive object dualities, behavioural sampling, reference changes and state coarse-graining down to the detector's resolution floor; they break only when the representation erases the latent channel or compresses the transition below that floor (Supplementary Table 3).

Emergence certificate. The frozen instruments are packaged into a single standardized adjudication (`emergence_certificat`
so that a new system receives a reproducible verdict: a three-condition qualification (regime-level: the collapse gate *and*, where the source analysis is available, joint-beyond-marginal reorganization; endogenous: the declared provenance boundary; persistent: no re-opening beyond 0.1 of reference within the observed window), the intensity profile as a vector (amplitude, abruptness class, sharpness, t^*, source ranking), and a categorical verdict with an

explicit *unresolvable* class below the measured power floor and a low-power flag between 12 and 40 grid points. We deliberately do not output a scalar "emergence score": amplitude and abruptness are demonstrated independent axes, and any fixed weighting of them would reintroduce the observable-of-convenience failure this framework exists to remove. Applied to the paper's systems, the certificate issues "emergent: punctuated" for the grip, convention, role and $N{=}100$ ant systems; the single-ant control—with nearly the same amplitude (0.92) as the colony—correctly *fails* qualification, because its collapse is purely individual-source rather than a joint regime, and the Overcooked occupancy object falls below the collapse gate with a low-power flag. Same entropy drop, different verdicts: the discrimination is exactly what a bare entropy measure cannot provide.

Learned systems. The grip-transport flagship uses 16 agents, grip threshold 6, a two-phase grip-then-push dynamics (grip gain 0.06, decay 0.01), trained by REINFORCE[38] (1,200 updates, batch 512, learning rate 2×10^{-3}) and, for the robustness check, by advantage actor–critic[39] on identical environment parameters (per-seed statistics in Supplementary Table 4). The hidden-coordination flagship uses 8 individually-observing agents with per-agent stances; the sticky variant sets stance-change probability 0.25 and the matched control sets it to 1.0 with all other constants imported unchanged. Overcooked runs use the standard cramped-room layout[51]. The non-constructed convention system is a ten-agent Lewis signalling population[40] (five meanings, five symbols, tabular policies, REINFORCE, 512 random speaker–listener pairs per update, 4,000 updates); the role system is six agents choosing among six roles with team reward only for full coverage (6,000 updates, otherwise identical training); in both, openness is computed from policy probabilities at 25-update checkpoints and adjudicated by the frozen detector (Supplementary Table 5). The neural replications keep both environments and recipes fixed and replace the tables with one-hidden-layer networks (32 tanh units, Adam 3×10^{-3}, ten fresh seeds per system): per-agent speaker and listener MLPs for the signalling population, and a single MLP shared by all six role agents, so that role differentiation must emerge inside one network. Because random initialization compresses commitment into the first few hundred updates, the neural runs are evaluated at 5-update resolution—a preregistered measurement-density amendment justified by the detector's validated grid-power curve; at the original 25-update grid the onsets are under-resolved (1/7 and 1/10), which is recorded alongside the fine-grid result (Supplementary Table 6). The initialization sweep varies the input-weight scale $\sigma \in \{0.02, 0.1, 0.45\}$ at five seeds per cell. The coordination-ring study trains standard `overcooked_ai` self-play PPO (mechanics and budget identical to the cramped-room runs, 2M steps) on the official `coordination_ring` layout, eight seeds, checkpoints every 20k steps; the regime object is the entropy of the pair's circulation direction (net winding around the central counter, Laplace-smoothed over 30 evaluation episodes per checkpoint), declared before running (per-seed conventions in Supplementary Table 7). A parallel preregistered attempt on the unmodified MPE `simple_spread` benchmark did not reach its frozen competence precondition (conflict-episode coverage 3% after 3,000 PPO updates) and is recorded as a training failure with no theory claim made either way. The causal commitment test perturbs stored ring checkpoints at 100k steps, at the seed's most-open checkpoint (last with direction probability in $[0.3, 0.7]$ and ≥ 20 committed evaluation episodes), and at 1.6M steps, adding per-tensor Gaussian noise of 0.25 or 0.5 relative standard deviations, resuming training for exactly 400k steps with unchanged mechanics and schedule position, and scoring the final convention against the seed's unperturbed direction; all times, scales, margins and clauses were frozen in the protocol file before any run. Because the two noise scales share a training seed, continuation runs cluster within seeds; the run-level Fisher test was the frozen clause, and a stricter seed-level paired analysis (exact sign-flip permutation over the eight seeds, labeled post-hoc in the protocol file) confirms the primary endpoint and demotes the strict direction-reversal contrast to descriptive. Because open and late checkpoints differ in training time as well as openness, the timing contrast alone cannot exclude generic training maturity; the fixed-time analysis that rules out the time confound exists in the grip system and has not yet been run here. The finite-size study uses sizes $N \in \{1, 2, 5, 10, 20, 50, 100, 200, 500\}$, 30 episodes per size and a 1,500-trip horizon, with t_{50} and width read from median openness crossings and the log-law slope bootstrapped over episodes (Supplementary Table 8). The ant, Kuramoto, Potts, Vicsek, Swift–Hohenberg and Schelling models follow their standard definitions (Supplementary

Methods).

Barrier measurement. The joint-exploration-barrier claim is tested by deterministic replay of both learned positive systems with policy snapshots at every checkpoint (replay validity: all ten re-run seeds reproduce the stored final codes and role assignments exactly). At the checkpoint nearest $t^*/2$ (onset seeds only), the unilateral adoption gain of agent i is the payoff of a probe hard-committed to the best of all $K!$ codes (respectively all R role locks) against the unmodified population, minus the payoff of i's own current policy, averaged over agents; payoff is symmetric expected intelligibility with a random partner (convention) or the permanent of the row-stochastic role matrix (roles). At the final checkpoint the unilateral deviation cost is the payoff on the population's majority regime minus the best payoff over all alternative committed regimes. Cross-play pairs speaker populations with listener populations from different seeds (20 ordered pairs) and assembles hybrid role teams from every proper subset split of every distinct-assignment seed pair (620 teams). All four outcome clauses were frozen before the run; one threshold-calibration miss (the convention deviation-cost clause set 0.50 where the payoff structure caps the cost at $0.4\times$within-population payoff for $K=5$ permutation codes, the measured value 0.3996 sitting at that ceiling) is reported verbatim in the protocol file. What these quantities measure is a *unilateral coordination barrier*—the absence of single-agent payoff gradient toward any regime, and the single-agent cost of leaving one—not the full performance landscape over joint policy paths; multi-agent coordinated escape routes are not enumerated.

Scalability. The pipeline's cost is dominated by openness estimation, not by the detector (which fits two linear models per gridpoint and is linear in grid length). Openness requires the entropy of the declared possibility object, estimated either exactly from policy probabilities (learned tabular and neural systems), or from empirical distributions over the declared coarse-graining (physical systems), whose sample complexity we measured directly: in the finite-sample audit the median absolute error of every ladder component is $\leq$0.02 bits at 3×10^4 samples, rising at 300 samples to 0.09 bits (individual), 0.31 (higher-order) and 0.81 (pairwise), so the pairwise term is the first casualty of scarce data. Total openness itself is linear in population size times horizon, and the breakpoint stage ran unchanged at population sizes up to $N=500$ in the finite-size study; the source decomposition is the only superlinear stage (pairwise marginals cost $O(N^2)$ and the higher-order maximum-entropy fit is the practical ceiling), so qualification and profile can always be computed at scales where the full typology cannot. All experiments in this paper run on CPU except the Overcooked PPO training; no single run exceeds a day of wall-clock time.

Preregistration and integrity. Every predicted outcome was frozen, with an explicit falsification clause, in an internal protocol file before the corresponding run; passes and misses are reported identically, and each amendment is recorded in place with its trigger (the complete ledger is collected in Supplementary Note 1). We characterize these as internally time-stamped frozen protocols rather than third-party registrations: they were not lodged with an external registry, and we release them verbatim—including every miss, every failed instrument extension and every post-hoc analysis labeled as such—together with the code, raw outputs and an output manifest, so that the ordering of protocol and result can be audited from the released record. All numbers in the main text and figures are produced by a single extraction script run directly against the raw output files; no value is transcribed by hand. One float32 numerical defect (an entropy clamp that rounded to a degenerate value) was found, fixed in code, and both affected experiments were rerun with identical seeds, with the corrupted outputs archived. No thresholds were changed after seeing results.

Reporting summary. Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article. The use of large-language-model agents is disclosed in the dedicated section in the back matter.

Data availability

The raw output of every experiment (one JSON file per registered run), together with the frozen protocol files that fix each prediction and falsification clause before running, is provided in the accompanying repository and will be archived with a permanent DOI on Zenodo upon acceptance. Source data for each main-text figure are generated directly from these files by the plotting script; no intermediate hand-edited data are used.

Code availability

All simulation, training, analysis and figure-generation code is available in the accompanying repository and will be archived on Zenodo upon acceptance. The manuscript's quantitative claims are reproduced by a single extraction script (`manuscript_numbers.py`) that reads the raw outputs and prints every number cited in the text and figures, by a figure-generation script (`make_figures.py`) that draws all panels from the same files, and by `make_si_tables.py`, which generates every Supplementary Table directly from the raw outputs.

Acknowledgements

Acknowledgements withheld for double-anonymized review.

Author contributions

Author contributions withheld for double-anonymized review.

Competing interests

The authors declare no competing interests.

Use of large language models

Large-language-model agents were used throughout this work, under continuous human direction and review, to design and implement experiments, refine analysis protocols, write and debug code, and draft and edit text. All predicted outcomes, falsification clauses and adjudication thresholds were fixed in the protocol files before the runs they govern; every number reported in the text and figures is generated by the released code from raw experiment outputs and cross-checked by an automated extraction script. The authors independently reviewed and validated the code, protocols and analyses, and take full responsibility for all claims.

References

[1] Philip W. Anderson. More is different. *Science*, 177(4047):393–396, 1972.

[2] Nigel Goldenfeld and Leo P. Kadanoff. Simple lessons from complexity. *Science*, 284(5411):87–89, 1999.

[3] Mark A. Bedau. Weak emergence. *Philosophical Perspectives*, 11:375–399, 1997.

[4] John H. Holland. *Emergence: From Chaos to Order*. Addison-Wesley, 1998.

[5] H. Eugene Stanley. *Introduction to Phase Transitions and Critical Phenomena*. Oxford University Press, 1971.

[6] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the 1/f noise. *Physical Review Letters*, 59(4):381–384, 1987.

[7] Scott Camazine, Jean-Louis Deneubourg, Nigel R. Franks, James Sneyd, Guy Theraulaz, and Eric Bonabeau. *Self-Organization in Biological Systems*. Princeton University Press, 2001.

[8] Iain D. Couzin, Jens Krause, Nigel R. Franks, and Simon A. Levin. Effective leadership and decision-making in animal groups on the move. *Nature*, 433(7025):513–516, 2005.

[9] David J. T. Sumpter. The principles of collective animal behaviour. *Philosophical Transactions of the Royal Society B*, 361(1465):5–22, 2006.

[10] Michele Ballerini, Nicola Cabibbo, Raphael Candelier, Andrea Cavagna, Evaristo Cisbani, Irene Giardina, Vivien Lecomte, Alberto Orlandi, Giorgio Parisi, Andrea Procaccini, Massimiliano Viale, and Vladimir Zdravkovic. Interaction ruling animal collective behavior depends on topological rather than metric distance: Evidence from a field study. *Proceedings of the National Academy of Sciences*, 105(4):1232–1237, 2008.

[11] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022.

[12] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020.

[13] Bowen Baker, Ingmar Kanitscheider, Todor Markov, Yi Wu, Glenn Powell, Bob McGrew, and Igor Mordatch. Emergent tool use from multi-agent autocurricula. In *International Conference on Learning Representations*, 2020.

[14] Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. Multi-agent reinforcement learning in sequential social dilemmas. In *International Conference on Autonomous Agents and Multiagent Systems*, pages 464–473, 2017.

[15] Max Jaderberg, Wojciech M. Czarnecki, Iain Dunning, Luke Marris, Guy Lever, Antonio Garcia Castaneda, Charles Beattie, et al. Human-level performance in 3d multiplayer games with population-based reinforcement learning. *Science*, 364(6443):859–865, 2019.

[16] Erik P. Hoel, Larissa Albantakis, and Giulio Tononi. Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49):19790–19795, 2013.

[17] Erik P. Hoel. When the map is better than the territory. *Entropy*, 19(5):188, 2017.

[18] Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461, 2016.

[19] Fernando E. Rosas, Pedro A. M. Mediano, Henrik J. Jensen, Anil K. Seth, Adam B. Barrett, Robin L. Carhart-Harris, and Daniel Bor. Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data. *PLoS Computational Biology*, 16(12):e1008289, 2020.

[20] Anil K. Seth. Measuring autonomy and emergence via granger causality. *Artificial Life*, 16(2): 179–196, 2010.

[21] Adam B. Barrett and Anil K. Seth. Practical measures of integrated information for time-series data. *PLoS Computational Biology*, 7(1):e1001052, 2011.

[22] Paul L. Williams and Randall D. Beer. Nonnegative decomposition of multivariate information. *arXiv preprint arXiv:1004.2515*, 2010.

[23] Pedro A. M. Mediano, Fernando E. Rosas, Andrea I. Luppi, Henrik J. Jensen, Anil K. Seth, Adam B. Barrett, Robin L. Carhart-Harris, and Daniel Bor. Greater than the parts: a review of the information decomposition approach to causal emergence. *Philosophical Transactions of the Royal Society A*, 380(2227):20210246, 2022.

[24] Steven H. Strogatz. From kuramoto to crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1-4):1–20, 2000.

[25] James P. Crutchfield. The calculi of emergence: computation, dynamics and induction. *Physica D: Nonlinear Phenomena*, 75(1-3):11–54, 1994.

[26] Melanie Mitchell. *Complexity: A Guided Tour.* Oxford University Press, 2009.

[27] Deep Ganguli, Danny Hernandez, Liane Lovitt, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova Dassarma, Dawn Drain, Nelson Elhage, et al. Predictability and surprise in large generative models. In *ACM Conference on Fairness, Accountability, and Transparency*, pages 1747–1764, 2022.

[28] Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*, 2023.

[29] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[30] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[31] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.

[32] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023.

[33] Edmund M. A. Ronald, Moshe Sipper, and Mathieu S. Capcarrère. Design, observation, surprise! A test of emergence. *Artificial Life*, 5(3):225–239, 1999.

[34] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[35] Chris R. Reid, Matthew J. Lutz, Scott Powell, Albert B. Kao, Iain D. Couzin, and Simon Garnier. Army ants dynamically adjust living bridges in response to a cost–benefit trade-off. *Proceedings of the National Academy of Sciences*, 112(49):15113–15118, 2015.

[36] Elad Schneidman, Susanne Still, Michael J. Berry, and William Bialek. Network information and connected correlations. *Physical Review Letters*, 91(23):238701, 2003.

[37] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.

[38] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3-4):229–256, 1992.

[39] Volodymyr Mnih, Adrià Puigdomènech Badia, Mehdi Mirza, Alex Graves, Timothy Lillicrap, Tim Harley, David Silver, and Koray Kavukcuoglu. Asynchronous methods for deep reinforcement learning. In *International Conference on Machine Learning*, pages 1928–1937, 2016.

[40] David K. Lewis. *Convention: A Philosophical Study*. Harvard University Press, Cambridge, MA, 1969.

[41] Andrea Baronchelli, Maddalena Felici, Vittorio Loreto, Emanuele Caglioti, and Luc Steels. Sharp transition towards shared vocabularies in multi-agent systems. *Journal of Statistical Mechanics: Theory and Experiment*, 2006(06):P06014, 2006.

[42] Damon Centola and Andrea Baronchelli. The spontaneous emergence of conventions: An experimental study of cultural evolution. *Proceedings of the National Academy of Sciences*, 112(7): 1989–1994, 2015.

[43] Rosie Zhao, Tian Qin, David Alvarez-Melis, Sham M. Kakade, and Naomi Saphra. Random scaling of emergent capabilities. In *International Conference on Machine Learning*, 2026. arXiv:2502.17356.

[44] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000.

[45] Igor Mordatch and Pieter Abbeel. Emergence of grounded compositional language in multi-agent populations. In *AAAI Conference on Artificial Intelligence*, volume 32, 2018.

[46] Angeliki Lazaridou, Alexander Peysakhovich, and Marco Baroni. Multi-agent cooperation and the emergence of (natural) language. In *International Conference on Learning Representations*, 2017.

[47] Jakob N. Foerster, Yannis M. Assael, Nando de Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[48] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[49] Peter Sunehag, Guy Lever, Audrunas Gruslys, Wojciech M. Czarnecki, et al. Value-decomposition networks for cooperative multi-agent learning based on team reward. In *International Conference on Autonomous Agents and Multiagent Systems*, pages 2085–2087, 2018.

[50] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Qmix: Monotonic value function factorisation for deep multi-agent reinforcement learning. In *International Conference on Machine Learning*, pages 4295–4304, 2018.

[51] Micah Carroll, Rohin Shah, Mark K. Ho, Thomas L. Griffiths, Sanjit A. Seshia, Pieter Abbeel, and Anca Dragan. On the utility of learning about humans for human–ai coordination. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[52] Juan A. Acebrón, L. L. Bonilla, Conrad J. Pérez Vicente, Félix Ritort, and Renato Spigler. The kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1):137–185, 2005.

[53] Tamás Vicsek, András Czirók, Eshel Ben-Jacob, Inon Cohen, and Ofer Shochet. Novel type of phase transition in a system of self-driven particles. *Physical Review Letters*, 75(6):1226–1229, 1995.

[54] Mark C. Cross and Pierre C. Hohenberg. Pattern formation outside of equilibrium. *Reviews of Modern Physics*, 65(3):851–1112, 1993.

[55] Thomas C. Schelling. Dynamic models of segregation. *Journal of Mathematical Sociology*, 1(2): 143–186, 1971.

[56] Fa-Yueh Wu. The potts model. *Reviews of Modern Physics*, 54(1):235–268, 1982.

[57] Oriol Vinyals, Igor Babuschkin, Wojciech M. Czarnecki, et al. Grandmaster level in starcraft ii using multi-agent reinforcement learning. *Nature*, 575(7782):350–354, 2019.

[58] Nolan Bard, Jakob N. Foerster, Sarath Chandar, Neil Burch, Marc Lanctot, H. Francis Song, Emilio Parisotto, Vincent Dumoulin, Subhodeep Moitra, Edward Hughes, et al. The hanabi challenge: A new frontier for ai research. *Artificial Intelligence*, 280:103216, 2020.

[59] Christoph Riedl. Emergent coordination in multi-agent language models. In *International Conference on Learning Representations*, 2026. arXiv:2510.05174.

[60] Adam S. Shai, Sarah E. Marzen, Lucas Teixeira, Alexander Gietelink Oldenziel, and Paul M. Riechers. Transformers represent belief state geometry in their residual stream. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

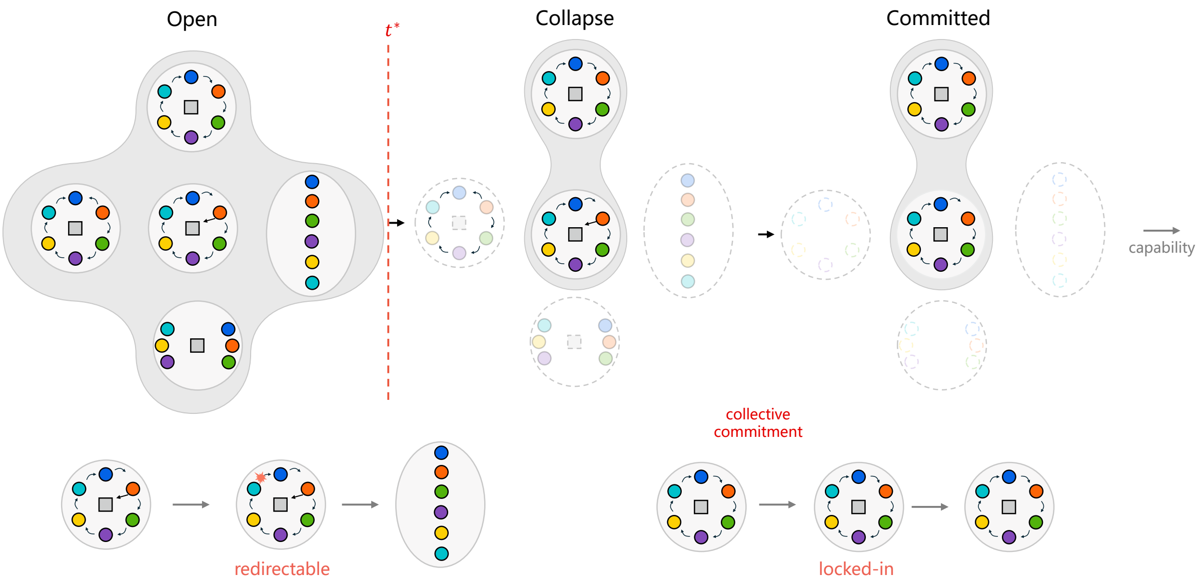


Figure 1. Emergence as possibility collapse. A population of learning agents (coloured dots) starts with many joint futures open (grey): several collective regimes—here a ring and a line—are still reachable. At a breakpoint t^* the joint possibility space begins to contract; alternatives fade until the population is committed to a single regime, and only after this collective commitment does the capability appear. Bottom, the practical consequence: while the space is open, a small perturbation (red arrow) can still redirect the population to a different regime; after commitment the same perturbation changes nothing—the regime is locked in.

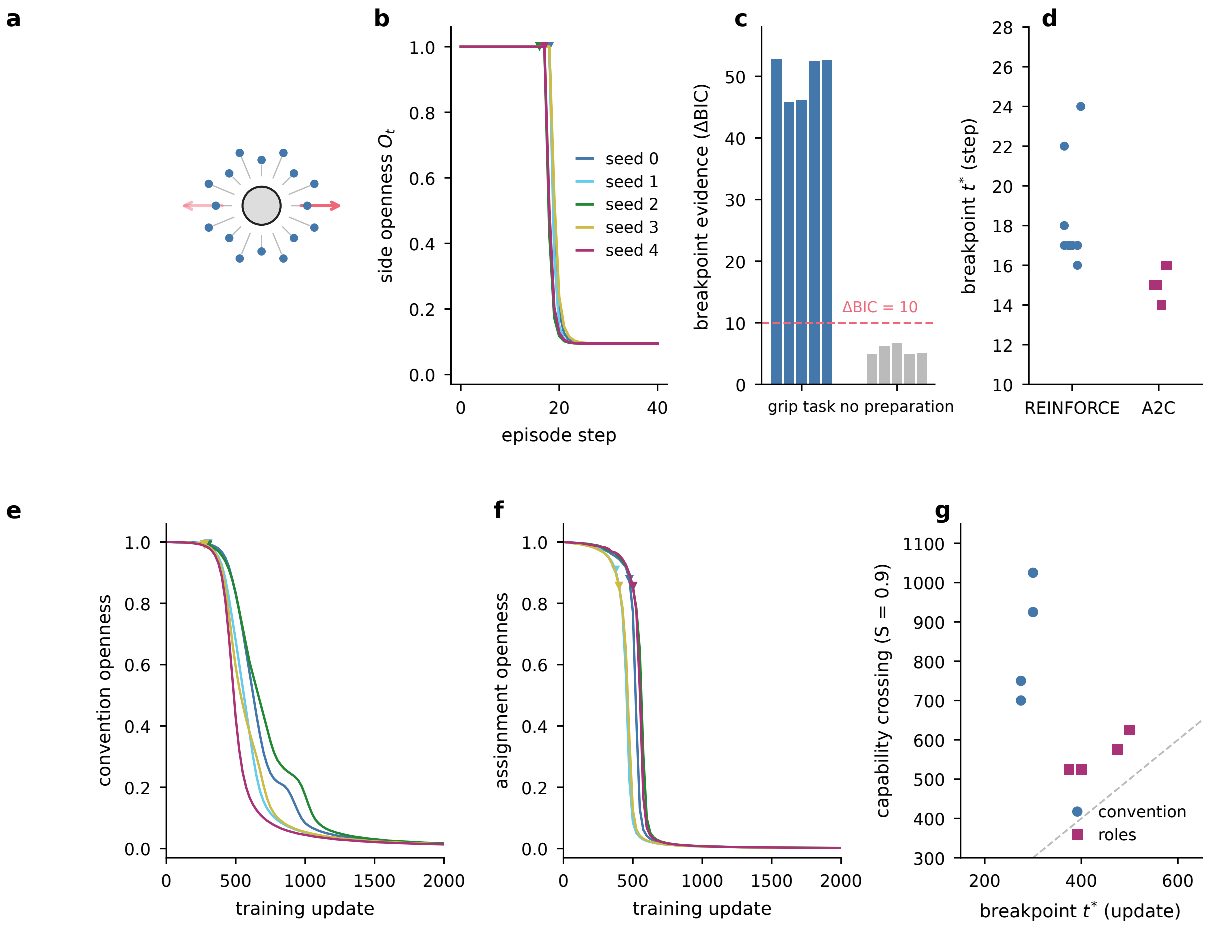


Figure 2. Punctuated realization in a learned collective. a, The grip-then-push transport task ($N = 16$): agents must jointly grip before the object can be pushed (arrows, the two possible push directions), structurally delaying the choice of side. **b**, Side-openness of the joint policy holds on a plateau and then collapses; triangles mark the fitted breakpoint t^* (5 REINFORCE seeds). **c**, Mechanism contrast: the grip task shows a breakpoint in 5/5 seeds (ΔBIC well above 10) whereas a matched no-preparation task shows none (0/5). **d**, Breakpoint location across seeds and algorithms: 10/10 for REINFORCE, and a strong partial replication under A2C (5/5 shape and primary hinge, 3/5 strict). **e**, Non-constructed positive I: a ten-agent signalling population with 120 exactly equivalent codes and no gate holds a flat open plateau, then commits with an onset breakpoint (4/5 seeds; triangles), each seed to a different code. **f**, Non-constructed positive II: six agents locking into a division of labour among 720 equivalent role permutations (5/5 onset, five distinct permutations). **g**, In all nine onset seeds the possibility-space breakpoint precedes the capability crossing ($S = 0.9$): commitment is measurable before the ability is expressed.

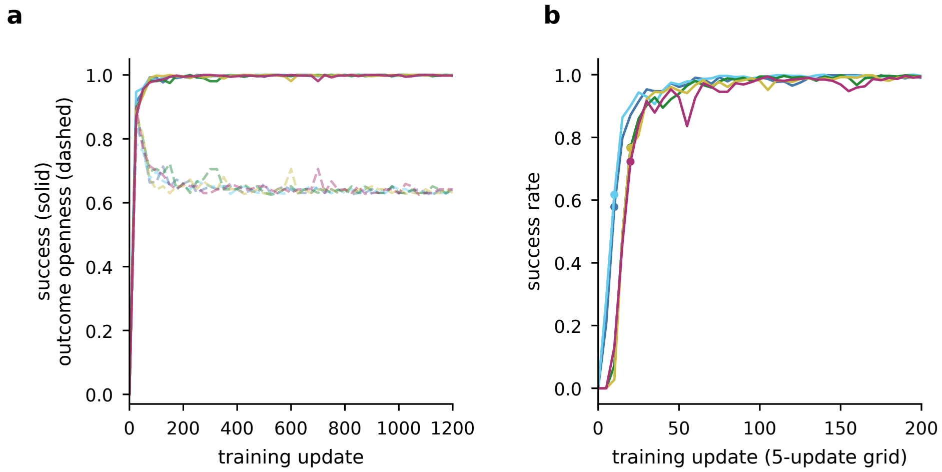


Figure 3. Two timescales dissociate in one learned system. a, Formation axis (across training): success (solid) and outcome openness (dashed) rise smoothly, no breakpoint (0/5). **b**, At fine 5-update resolution the success rise is fast (per-seed midpoints, dots, at updates 10–20) but still has no structural onset (0/5). Formation is smooth and expansive while realization within an episode is punctuated (Fig. 2b)—same system, same instrument.

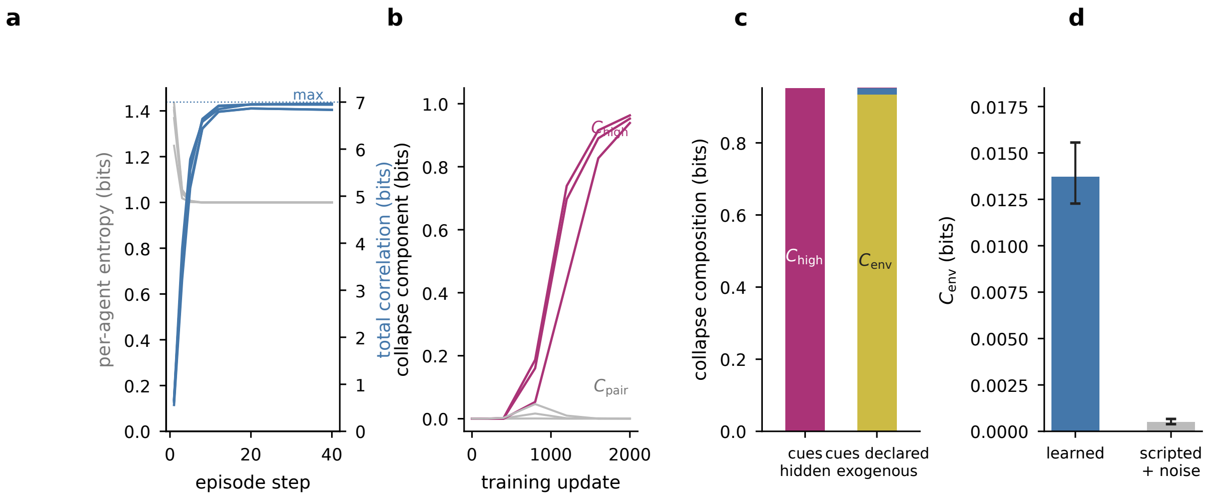


Figure 4. The source typology transfers to learning. a, Hidden-coordination task: per-agent entropy stays open while total correlation rises to near the 7-bit maximum—a purely relational collapse. **b**, With the low-order route blocked, learning builds an irreducible higher-order (XOR) carrier: C_{high} grows while C_{pair} stays near zero. **c**, Contract-relativity on the learned system: declaring the private cues exogenous re-attributes the identical collapse from C_{high} to C_{env}. **d**, At matched task score, the environment-conditioned component separates a learned pair from a noise-matched scripted pair (non-overlapping 95% confidence intervals).

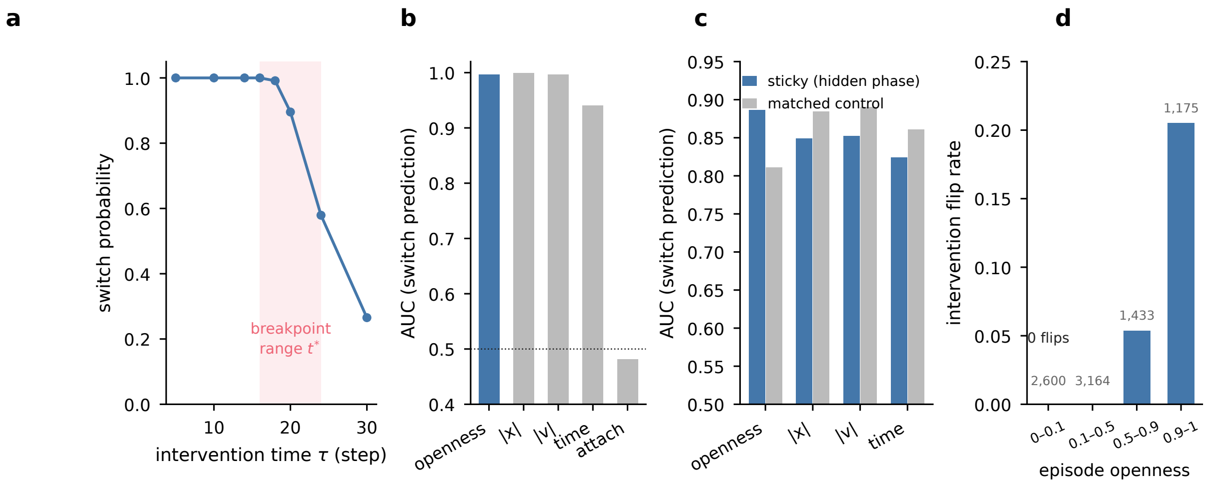


Figure 5. Remaining openness predicts controllability. a, Grip system: probability that a counter-regime impulse switches the outcome, as a function of intervention time; the window closes after the breakpoint range (shaded). **b**, Openness predicts switchability as well as or better than physical baselines (AUC). **c**, Matched-parameter causal contrast: openness beats the order parameter only when a hidden consolidation phase exists (sticky), and the ranking reverses in the control. **d**, Ant colony, paired counterfactuals: intervention flip-rate rises with the episode's own remaining openness; closed episodes never flip.

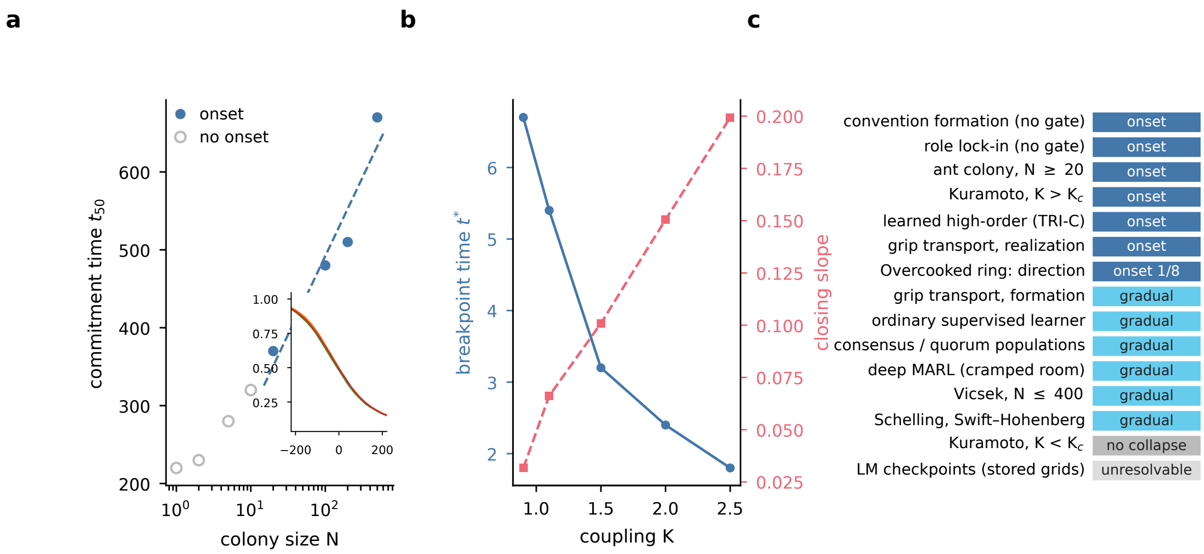
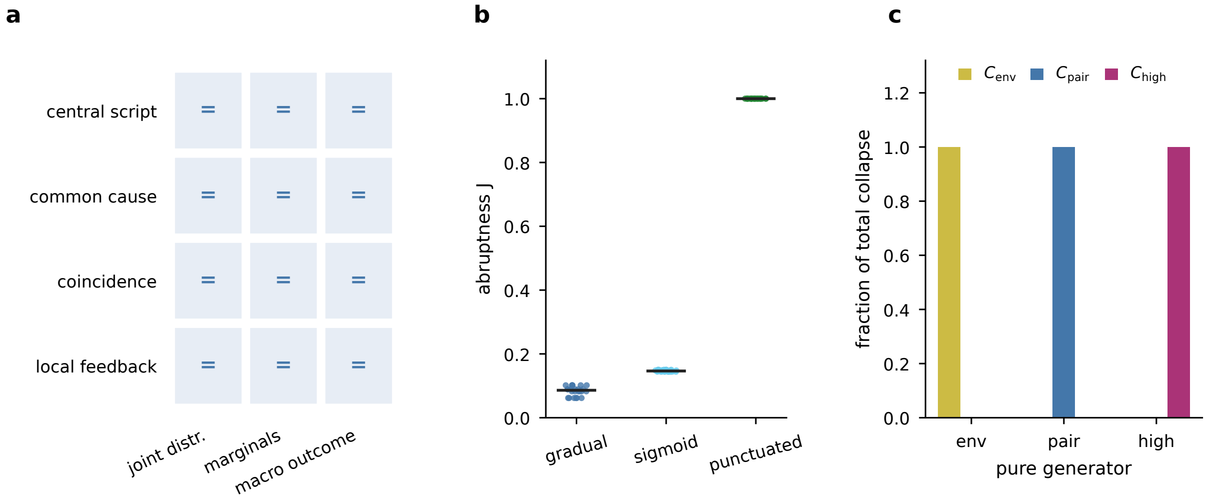


Figure 6. Scaling and scope. a, Finite-size scaling in the ant model: commitment time is consistent with $t_{50} = a + b \ln N$ ($b = 88$, 95% CI [41, 124], $R^2 = 0.93$; dashed fit), as predicted by nucleation from $N^{-1/2}$ fluctuations; open circles, sizes below the onset threshold. Inset: median collapse curves for $N = 50$–500 superpose under pure time translation (aligned RMS 0.010 vs 0.266 unaligned)—the profile is universal, only the timing shifts. **b**, Kuramoto criticality: breakpoint time falls and closing slope rises as coupling increases; subcritical runs are gated null. **c**, Classification: onset appears where a joint-regime barrier is crossed—including without any designed gate (convention formation, role lock-in) and in the standard Overcooked coordination-ring layout, where all 8 seeds commit to a circulation direction and 1/8 certifies onset (Results)—and is absent for gradual convergence (formation, ordinary learners, consensus/quorum populations, cramped-room deep MARL, several classic cases).



Extended Data Fig. 1. Instrument validation on constructed ground truth. **a**, Four generators—central script, common cause, coincidence and local feedback—constructed so that no statistic of the shared joint-action distribution separates them (identical joints, marginals and macroscopic success). What separates them is the declared contract: conditioning on declared exogenous variables, interaction-severing counterfactuals and persistence checks assign each to its predicted certificate component—attribution is contract-relative, not mechanism discovery from outcome statistics. **b**, Full-factorial calibration (72 cells): amplitude M is invariant across temporal shape while abruptness J strictly orders punctuated $>$ sigmoid $>$ gradual. **c**, Source-decomposition ladder on analytic ground truth: each pure generator loads almost entirely onto its own component.